

Assignment No - 1 (Basic square patterns)

Q. 1]

print an $n \times n$ star square (nested loops)

$n = 4$

→

```
* * * *
* * * *
* * * *
* * * *
```

→

```
class P1 {
```

```
    p.s.v.m (string args[]) {
```

```
        for (int i = 1; i ≤ 4; i++) {
```

```
            for (int j = 1; j ≤ 4; j++) {
```

```
                s.o.p ("* ")
```

```
            }
```

```
        s.o.p ();
```

```
    }
```

```
}
```

Dry run :-

$i = 1; \quad i \leq 4 \quad \text{True.}$

$j = 1 \quad j \leq 4 \quad \text{True.} \quad i = 1$

$s.o.p ("*");$

$j = 2 \quad j \leq 4 \quad \text{True}$

$s.o.p ("*");$

$j = 3 \quad j \leq 4 \quad \text{True}$

$s.o.p ("*");$

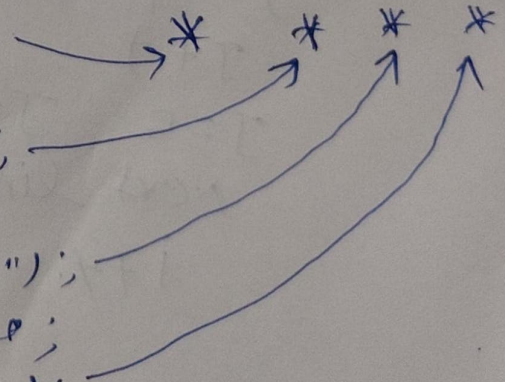
$j = 4 \quad j \leq 4 \quad \text{True};$

$s.o.p ("*");$

$j = 5 \quad j \leq 4 \quad \text{false};$

$i++$

1st line



$i = 2$ $i \leq 4$; True.

2nd line

$j = 1$ $J \leq 4$ True.

S.O.P (" * ");

$J++$

$J = 2$ $J \leq 4$ True;

S.O.P (" * ");

$J++$

$J = 3$ $J \leq 4$ True.

S.O.P (" * ");

$J++$

$J = 4$ $J \leq 4$ True.

S.O.P (" * ");

$J++$

$J = 5$ $J \leq 4$ False.

$i++$;
next line.

$i = 1 \rightarrow$ * * * *

$i = 2 \rightarrow$ * * * *

$i = 3$; $i \leq 4$; True.

$J = 1$ $J \leq 4$ True S.O.P (*);

$J++$

$J = 2$ $J \leq 4$ True S.O.P (*);

$J++$

$J = 3$ $J \leq 4$ True S.O.P (*);

$J++$

$J = 4$ $J \leq 4$ True S.O.P (*);

$J++$

$J = 5$ $J \leq 4$ False

Next line ;

$i++$.

$i = 1$ * * * *

$i = 2$ * * * *

$i = 3$ * * * *

3rd row

$i = 4$ $i \leq 4$ True,

4th row

$j = 1$: $j \leq 4$ True s.op (*); $\rightarrow$ * * * *

$j = 2$: $j \leq 4$ True s.op (*); $\rightarrow$ * * * *

$j = 3$: $j \leq 4$ True s.op (*); $\rightarrow$ * * * *

$j = 4$: $j \leq 4$ True s.op (*); $\rightarrow$ * * * *

$j = 5$: $j \leq 4$ False

Next line

$i++$

$i = 1$ * * * *

$i = 2$ * * * *

$i = 3$ * * * *

$i = 4$ * * * *

$i = 5$; $i \leq 4$ False,

Exit

Output :-

```
* * * *
* * * *
* * * *
* * * *
```


2) pattern

```
1 1 1 1
2 2 2 2
3 3 3 3
4 4 4 4
```

```
→ class pattern {
    p.s.v.m (string[] args) {
        for (int i=1; i≤4; i++) {
            for (int j=1; j≤4; j++) {
                s.o.p (i + " ");
            }
            s.o.pln ();
        }
    }
}
```

Dry run :- $i=1$ $i \leq 4$ True;

1st row

$j \leq 1$ $j \leq 4$ True

s.o.p (i); j++ →

$j=2$ $j \leq 4$ True.

s.o.p (i) j++ →

$j=3$ $j \leq 4$ True

s.o.p (i) j++ →

$j=4$ $j \leq 4$ True.

s.o.p (i) j++ →

$j=5$ $j \leq 4$ False.

next line;

i++;

$i=1 \rightarrow 1 \ 1 \ 1 \ 1$

$i = 2$ $j \leq 4$ True

2nd row

$j = 1$ $J \leq 4$ True,
s.o.p (i) $J++$

$j = 2$ $J \leq 4$ True,
s.o.p (i) $J++$

$J = 3$ $J \leq 4$ True,
s.o.p (i) $J++$

$J = 4$ $J \leq 4$ True
s.o.p (i) $j++$

$J = 5$ $J \leq 4$ False

Next line

3rd row $i++;$

$i = 3$ $j \leq 4$ True.

$j = 1$ $J \leq 4$ True; s.o.p (i); $j++;$

$j = 2$ $J \leq 4$ True; s.o.p (i); $j++;$

$j = 3$ $J \leq 4$ True; s.o.p (i); $j++;$

$j = 4$ $J \leq 4$ True; s.o.p (i); $j++;$

$J = 5$ $J \leq 4$ False. Next line

$i++;$

4th row

$i = 4$ $i \leq 4$ True;

$j = 1$ $J \leq 4$ True s.o.p (i) $j++;$

$J = 2$ $J \leq 4$ True s.o.p (i) $j++;$

$J = 3$ $J \leq 4$ True s.o.p (i) $j++;$

$J = 4$ $J \leq 4$ True s.o.p (i) $J++;$

$J = 5$ $J \leq 4$ False Next line.

$j++.$

$i \leq 5$ $i \leq 4$ False

$i = 1 \rightarrow 1 \ 1 \ 1 \ 1$
 $i = 2 \rightarrow 2 \ 2 \ 2 \ 2$

$i = 1 \rightarrow 1 \ 1 \ 1 \ 1$
 $i = 2 \rightarrow 2 \ 2 \ 2 \ 2$
 $i = 3 \rightarrow 3 \ 3 \ 3 \ 3$

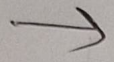
$i = 1 \rightarrow 1 \ 1 \ 1 \ 1$
 $i = 2 \rightarrow 2 \ 2 \ 2 \ 2$
 $i = 3 \rightarrow 3 \ 3 \ 3 \ 3$
 $i = 4 \rightarrow 4 \ 4 \ 4 \ 4$

output

1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4

3) pattern

```
1 2 3 4
1 2 3 4
1 2 3 4
1 2 3 4
```



Class pattern {

P.S.V.m (String, args[]) {

For (int i = 1 ; i ≤ 4 ; i++) {

For (int j = 1 ; j ≤ 4 ; j++) {

S.o.p (j + " ");

}

S.o.pln ();

}

1st row

dry run:- i = 1 @ i ≤ 4 True

j = 1 j ≤ 4 True. S.o.p(j); j++ → 1 2 3 4

j = 2 j ≤ 4 True S.o.p(j); j++ →

j = 3 j ≤ 4 True S.o.p(j); j++ →

j = 4 j ≤ 4 True S.o.p(j); j++.

j = 5 j ≤ 4 False

next line

i++

i = 1 → 1 2 3 4

2nd row

$i = 2$ $i \leq 4$ True;

↓

$j = 1$ $j \leq 4$ True; s.op(j) j++;

$j = 2$ $j \leq 4$ True; s.op(j) j++;

$j = 3$ $j \leq 4$ True; s.op(j) j++;

$j = 4$ $j \leq 4$ True; s.op(j) j++;

$j = 5$ $j \leq 4$ False; Next line.

i++

$i = 1 \rightarrow 1 \ 2 \ 3 \ 4$
 $i = 2 \rightarrow 1 \ 2 \ 3 \ 4$

3rd row

$i = 3$ $i \leq 4$ True;

$j = 1$ $j \leq 4$ True; s.op(j) j++;

$j = 2$ $j \leq 4$ True; s.op(j) j++;

$j = 3$ $j \leq 4$ True; s.op(j) j++;

$j = 4$ $j \leq 4$ True; s.op(j) j++;

$j = 5$ $j \leq 4$ False; Next line

i++

$i = 1 \rightarrow 1 \ 2 \ 3 \ 4$
 $i = 2 \rightarrow 1 \ 2 \ 3 \ 4$
 $i = 3 \rightarrow 1 \ 2 \ 3 \ 4$
 $i = 4 \rightarrow 1 \ 2 \ 3 \ 4$

4th row

$i = 4$ $i \leq 4$ True;

$j = 1$ $j \leq 4$ True; s.op(j) j++;

$j = 2$ $j \leq 4$ True; s.op(j) j++;

$j = 3$ $j \leq 4$ True; s.op(j) j++;

$j = 4$ $j \leq 4$ True; s.op(j) j++;

$j = 5$ $j \leq 4$ False; Next line

i++

$i = 1 \rightarrow 1 \ 2 \ 3 \ 4$
 $i = 2 \rightarrow 1 \ 2 \ 3 \ 4$
 $i = 3 \rightarrow 1 \ 2 \ 3 \ 4$
 $i = 4 \rightarrow 1 \ 2 \ 3 \ 4$

$i = 5$ $i \leq 4$ False

Exit.

Output:-

1	2	3	4
1	2	3	4
1	2	3	4
1	2	3	4